

(a) INPUTS**7 VARIABLES
+ MASK**

WTEMP • FLOW
TEMP • PRCP
RHMEAN • DH
WDSP

32-day buffer
construction only
≠ effective
memory

WLEVEL

excluded

ANCHOR A

frozen damped-
persistence anchor

fitted blend of
last y and
climatology

(b) REPRESENTATION**SPARSE ROUTER**

7 variables
lags 0–14

LEFT-LOOKING TCN

2 blocks • kernel 3
receptive field 7

**MIXTURE OF
EXPERTS**

combined
representation

PROPOSAL P

learned relaxation κ

(c) HEADS**EVENT HEAD**

event score

Q HEADS

q50: separate
anchor-bounded
q05/q95: $\pm$ widths

POINT HEAD

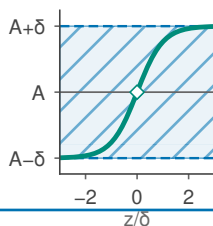
MSE residual r

(d) CALIBRATION**PLATT**

fit 2018 only

**CALIBRATED
PROBABILITY****CQR**

member average
fit 2018 only

**CALIBRATED
INTERVAL****ANCHOR-BOUNDED POINT IDENTITY**

$$\hat{y} = A + \delta \tanh(z/\delta)$$

$$z = P - A + r \cdot \delta = 1.0 \text{ } ^\circ\text{C}$$

Deviation from anchor;
not an error or safety bound

κ / router: internal allocations $\neq$ physical routing